

1. Image Enhancement in Spatial Domain:

It means improving the quality of an image.

In the spatial domain, we directly change the pixels of the image.

main methods:

1. Gray-Level transformation - changes image brightness.
2. Spatial filtering - removes noise or improves details.

ex: . Negative - changes dark areas to bright
mean/median filter → removes noise.

∴ Finally, the image enhancement makes an image clearer, brighter & easier to understand.

2. Smoothing and ~~Sharpening~~

It means removing of noise from an image.

Three types of filters:

1. mean
2. median
3. Gaussian

uses:- Removes unwanted noise & makes the image smooth.

Sharpening: It means making edges & details clearer.

Filters $\rightarrow$ Laplacian
 $\rightarrow$ Sobel

uses: makes blurred images & object edges clearer.

3 Frequency Domain Enhancement:-

Improves an image by using frequency components.

Steps:

Image $\rightarrow$ Fourier Transform $\rightarrow$ filter $\rightarrow$ Enhanced image.

• Low-Pass Filter:

• Removes high frequencies

• Reduces noise.

• makes image smooth.

• High-Pass Filter:

• Removes low frequencies

• Highlights edges

• makes image sharp.

Image Segmentation:

means dividing an image into different of meaningful parts.

Types:

1. point detection: It finds individual points.

Use: finding the small defects

2. Line Detection: It finds Lines

Use: Road Lines

3. Edge Detection: Finds object edges.

Use: object recognition.

4. Thresholding: Separates the object from the background using brightness.

Use: Object extraction.

5. Edge Detection, Edge Linking and Boundary detection.

• Edge Detection: Finds the edges of objects
ex: Sobel.

• Edge Linking: Connects broken edges together.

• Boundary detection:

Finds the complete boundary of an object